

5.1 Verify that if μ_1 and μ_2 are finite measure on $(\mathcal{R}_1, \mathcal{F}_1)$ and $(\mathcal{R}_2, \mathcal{F}_2)$, then $\mu_{12}(\cdot)$ and $\mu_{21}(\cdot)$, defined by (1.4) and (1.5) are measures on $(\mathcal{R}_1 \times \mathcal{R}_2, \mathcal{F}_1 \times \mathcal{F}_2)$

$$\mu_{12}(A) = \int_{\mathcal{R}_1} \mu_2(A|w_1) \mu_1(dw_1) \quad (w_1 \in \mathcal{R}_1, A|w_1 \in \mathcal{F}_2) \quad A \in \mathcal{F}_1 \times \mathcal{F}_2$$

$$\mu_{21}(A) = \int_{\mathcal{R}_2} \mu_2(A|w_2) \mu_2(dw_2)$$

$\forall A \in \mathcal{F}_1 \times \mathcal{F}_2$, $\mu_{12}(\cdot)$ is defined by Lebesgue integral, so $\mu_{12}(A) \geq 0$

$$\mu_{12}(\phi) = \int_{\mathcal{R}_1} \mu_2(\phi) d\mu_1(\phi) = 0$$

To prove the countable additivity of $\mu_{12}(\cdot)$.

For a disjoint sequence $\{A_n\}_{n=1}^{\infty}$ in $\mathcal{F}_1 \times \mathcal{F}_2$ s.t. $B = \bigcup_{n=1}^{\infty} A_n$

$$\text{then } \mu_{12}(B) = \mu_{12}\left(\bigcup_{n=1}^{\infty} A_n\right) = \int_{\mathcal{R}_1} \mu_2(B|w_1) d\mu_1(w_1) \quad B|w_1 = \{w_2 \in \mathcal{R}_2 \mid (w_1, w_2) \in B\}.$$

$$\text{For any fixed } w_1, B|w_1 = \bigcup_{n=1}^{\infty} (A_n)|w_1.$$

$$\Rightarrow \mu_2(B|w_1) = \sum_{n=1}^{\infty} \mu_2((A_n)|w_1)$$

$$\text{By MCT} \quad \int_{\mathcal{R}_1} \mu_2(B|w_1) d\mu_1(w_1) = \sum_{n=1}^{\infty} \int_{\mathcal{R}_1} \mu_2((A_n)|w_1) d\mu_1(w_1) = \sum_{n=1}^{\infty} \mu_{12}(A_n)$$

$$\Rightarrow \mu_{12}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mu_{12}(B) = \sum_{n=1}^{\infty} \mu_{12}(A_n)$$

So, $\mu_{12}(\cdot)$ is a measure on $\mathcal{F}_1 \times \mathcal{F}_2$. Similarly, $\mu_{21}(\cdot)$ is also measure on $\mathcal{F}_1 \times \mathcal{F}_2$.

5.6 Let $(\mathcal{R}, \mathcal{F}, \mu)$ be a σ -finite measure space and f be a nonnegative measurable function. Then $\int_{\mathcal{R}} f d\mu = \int_{[0, \infty)} \mu(\{f \geq t\}) dt$.

Define $g(w, t) = \begin{cases} 1 & f(w) \geq t \\ 0 & f(w) < t \end{cases}$, $g(w, t)$ is $\mathcal{F} \times \mathcal{B}(\mathbb{R}^+)$ -measurable.

$$\int_{\mathcal{R}} g(w, t) d\mu = \mu(\{f(w) \geq t\}) \text{ and } \int_{\mathbb{R}^+} g(w, t) dt = f(w)$$

Apply Tonelli's Theorem, we will get $\int_{\mathcal{R}} f d\mu = \int_{\mathbb{R}^+} \mu(\{f \geq t\}) dt$.

5.10 Show that $I \equiv \int_0^\infty e^{-x^2/2} dx = \sqrt{\frac{\pi}{2}}$

$$I = \int_0^\infty e^{-x^2/2} dx = \int_0^\infty e^{-y^2/2} dy$$

$$\begin{aligned} I^2 &= \int_0^\infty e^{-x^2/2} dx \cdot \int_0^\infty e^{-y^2/2} dy \\ &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)/2} dx dy = \int_0^{\pi/2} \int_0^\infty e^{-\frac{r^2}{2}} r dr d\theta \\ &= \int_0^\infty r e^{-\frac{r^2}{2}} dr \cdot \int_0^{\pi/2} 1 d\theta = \frac{\pi}{2} \end{aligned}$$

$$\Rightarrow I = \sqrt{\frac{\pi}{2}}.$$

5.11 Let μ be a finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let, $f, g: \mathbb{R} \rightarrow \mathbb{R}^+$ be nondecreasing.

Show that $\mu(\mathbb{R}) \int fg d\mu \geq (\int f d\mu) (\int g d\mu)$

$$h(x_1, x_2) = (f(x_1) - f(x_2))(g(x_1) - g(x_2)) \text{ on } \mathbb{R}^2.$$

$$\int_{\mathbb{R}^2} h(x_1, x_2) d\mu^2 \geq 0 \Rightarrow \int_{\mathbb{R}^2} (f(x_1) - f(x_2))(g(x_1) - g(x_2)) d\mu^2 = \int_{\mathbb{R}^2} f(x_1)g(x_1) - f(x_1)g(x_2) - f(x_2)g(x_1) + f(x_2)g(x_2) d\mu^2$$

apply Tonelli's Theorem.

$$\begin{aligned} \int_{\mathbb{R}^2} h(x_1, x_2) d\mu^2 &= \int_{\mathbb{R}^2} f(x_1)g(x_1) - f(x_1)g(x_2) - f(x_2)g(x_1) + f(x_2)g(x_2) d\mu^2 = 2 \int_{\mathbb{R}} f g d\mu - \int_{\mathbb{R}^2} f(x_1)g(x_2) d\mu(x_1) d\mu(x_2) \\ &\quad - \int_{\mathbb{R}^2} f(x_2)g(x_1) d\mu(x_2) d\mu(x_1) \geq 0 \end{aligned}$$

$$\Rightarrow \mu(\mathbb{R}) \int_{\mathbb{R}} f g d\mu \geq \int g d\mu \int f d\mu.$$

Problem A. $\Omega_i = \mathcal{N}$, $\mathcal{F}_i = \mathcal{P}(\mathcal{N})$. $\mu_i = \text{counting measure}$. $i=1,2$.

Define $f: (\mathcal{N}_1 \times \mathcal{N}_2) \rightarrow \mathbb{R}$ as $f(w_1, w_2) = I_{\{(n, n): n \in \mathcal{N}\}}(w_1, w_2) - I_{\{(n, n+1): n \in \mathcal{N}\}}(w_1, w_2)$.

(i) Prove $\mathcal{F}_1 \times \mathcal{F}_2 = \mathcal{P}(\mathcal{N} \times \mathcal{N})$

(ii) Is f a $\mathcal{F}_1 \times \mathcal{F}_2$ measurable function?

(iii) Is f integrable w.r.t. $\mu_1 \times \mu_2$

(iv) Compute. $\int_{\mathcal{N}_1} [\int_{\mathcal{N}_2} f(w_1, w_2) d\mu(w_2)] d\mu(w_1)$
 $\int_{\mathcal{N}_2} [\int_{\mathcal{N}_1} f(w_1, w_2) d\mu(w_1)] d\mu(w_2)$

(i) Let C be $\{A \times B \mid A \in \mathcal{F}_1, B \in \mathcal{F}_2\}$ by definition $\mathcal{F}_1 \times \mathcal{F}_2 = \sigma(C)$.

$$C \subset \mathcal{P}(\mathcal{N} \times \mathcal{N}) \Rightarrow \mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{P}(\mathcal{N} \times \mathcal{N}).$$

$$\forall D \in \mathcal{P}(\mathcal{N} \times \mathcal{N}) \text{ s.t. } D = \bigcup_{i=1}^\infty \{(a_i, b_i)\} \quad \forall a_i, b_i \in \mathcal{N}.$$

$$\Rightarrow D \in \mathcal{G} \Rightarrow P(\mathcal{U} \times \mathcal{V}) \subset \sigma(\mathcal{G}) = \mathcal{F}_1 \times \mathcal{F}_2$$

$$\text{So } \mathcal{F}_1 \times \mathcal{F}_2 = P(\mathcal{U} \times \mathcal{V}).$$

(ii) both $\{(n, n), n \in \mathcal{U}\}$ and $\{(n, n+1), n \in \mathcal{U}\}$ are in $\mathcal{F}_1 \times \mathcal{F}_2$

$\Rightarrow f$ is a $\mathcal{F}_1 \times \mathcal{F}_2$ measurable function.

$$(iii) \int_{\mathcal{U}_1 \times \mathcal{U}_2} |f(w_1, w_2)| d\mu_1 \times \mu_2 = \sum_{n=1}^{\infty} \mu_1 \times \mu_2(\{(n, n)\}) + \sum_{n=1}^{\infty} \mu_1 \times \mu_2(\{(n, n+1), n \in \mathcal{U}\}) = \infty.$$

So f is not integrable w.r.t. $\mu_1 \times \mu_2$.

(iv) For each $w_1 \in \mathcal{U}$.

$$\int_{\mathcal{U}_2} f(w_1, w_2) d\mu_2(w_2) = [I_{\{(n, n), n \in \mathcal{U}\}}(w_1, w_1) - I_{\{(n, n+1), n \in \mathcal{U}\}}(w_1, w_1+1)] = 0$$

$$\text{So } \int_{\mathcal{U}_1} \left[\int_{\mathcal{U}_2} f(w_1, w_2) d\mu_2(w_2) \right] d\mu_1(w_1) = 0$$

Similarly, $\int_{\mathcal{U}_1} f(w_1, 1) d\mu_1(w_1) = 1$ and $\int_{\mathcal{U}_1} f(w_1, w_2) d\mu_1(w_1) = 0$ for all $w_2 \geq 2$.

$$\Rightarrow \int_{\mathcal{U}_2} \left[\int_{\mathcal{U}_1} f(w_1, w_2) d\mu_1(w_1) \right] d\mu_2(w_2) = \mu_2(\{1\}) = 1.$$

Problem B. Let $f = 1_A$, $g = 1_B$ be two indicator functions for sets $A, B \in \mathcal{B}(\mathbb{R})$.

Express the convolution function $(f * g)(x)$ in terms of Lebesgue measure of sets, for each $x \in \mathbb{R}$.

$$(f * g)(x) = \int \int I_S(x+y) \cdot I_A(x) dm \cdot I_B(y) dm \quad \text{m is the Lebesgue measure on } \mathbb{R}.$$